

WHAT HELPS PCB DEFECT CLASSIFICATION: CROP SCALE, NOT GENERATIVE AUGMENTATION

Edward K.Y. Yapp

School of Business, Singapore University of Social Sciences, Singapore

edwardyappky@suss.edu.sg

Keywords: PCB, DEFECT CLASSIFICATION, GENERATIVE AUGMENTATION, DATA SCARCITY, VQ-VAE-2

Abstract

Synthetic defect images are widely proposed to address the scarcity of defective samples in printed circuit board (PCB) inspection. We test that claim quantitatively. Scarcity is simulated by restricting a classifier to 10, 25, 50 or 100% of the available crops; at each level a class-conditional VQ-VAE-2 is trained on that same restricted data and its samples are added to the classifier’s training set. Adding them changes nothing at 10% (-0.008 macro-F1 against real data alone; 95% CI $[-0.031, +0.014]$, ten seeds) and lowers mean performance at 25, 50 and 100%; only at 25% is the decrease significant. Allowing the generator to train on all available crops, while the classifier still sees only its 215 (the 10% level), also changes nothing ($+0.002$), so the limit is not the generator’s own data. Replacing generated crops with autoencoder reconstructions of real ones—same encoder and decoder, but no generative step—gives $+0.51$, a large gain that the generator does not deliver. A simpler defect/no-defect variant shows no benefit either. By contrast, cropping to the defect instead of using the released 600-pixel window raises the same classifier from 0.25 to 0.90 macro-F1: what matters is crop scale and not generative augmentation.

1 Introduction

Defective PCB samples are scarce because production is efficient, and generative models have been proposed as a source of synthetic training data [1], including in our own previous work [2]. That work generated PCB defect images with VQ-VAE-2 [3] and argued qualitatively that it addresses scarcity, but did not test whether the generated images improve a downstream task. This paper reports that test.

We compare three sources of added training data at each level of simulated scarcity, and separately compare the released 600-pixel crop protocol against crops centred on the defect.

2 Experimental setup

2.1 Data

The PCB dataset of [4] contains 10 board designs with six defect classes injected onto shared clean templates (the ‘real’ dataset is thus itself synthetic). From the released 600×600 crops [5], where the median defect occupies only 27×27 pixels (about 0.2% of the frame), we cut one tight crop per annotated bounding box (128×128 native, resized to 256×256), centred on the defect nearest the frame centre where a crop holds several, so each image is dominated by one defect. Two boards are held out entirely for testing (518 crops), so no board appears in both train and test; the training pool holds 2,149 primary crops from the other eight, and nested subsamples give levels of 10/25/50/100% (215/537/1,075/2,149).

2.2 Generator

We use VQ-VAE-2 with PixelSNAIL priors [3]. As in [3] (Fig. 2b), the top-level prior is conditioned on the class label and the bottom-level prior on the class label as well as the top-level

code. The public implementation we build on (rosinality/vq-vae-2-pytorch) targets a single-class dataset and omits class conditioning, so we add it. A per-class embedding enters at three places: added to the input feature map, injected through the conditioning stack and added to the output logits. As a control, we train a reduced-capacity copy of the top prior to memorise six crops, one per class. Sampling by label, it matches its own class’s crop 99.1% of the time against 19.2% for the others, confirming the label reaches the output. Under the restricted setting the VQ-VAE-2 and both priors are trained from scratch on the restricted data available at that level (at 100%, on all annotated defects within those crops); under the full-pool setting the generator sees all available crops while the classifier remains restricted.

2.3 Downstream task

An ImageNet-pretrained ResNet-18 is trained at each level of scarcity under three conditions: (A) real crops only; (B) plus synthetic crops; (C) plus synthetic crops filtered by a hard-agreement variant of the classifier-based rejection sampling of [3], keeping only those a classifier assigns to the class they were conditioned on. That filter is the same classifier as condition A at the same level, so it cannot pass information from the withheld data into the filtered set.

3 Results

3.1 Crop scale dominates

At full data, the released 600-pixel protocol reaches only 0.246 ± 0.013 macro-F1, because a 27-pixel defect shrinks to about 10 pixels once the crop is resized. Cutting one tight crop per 600-pixel crop leaves the dataset size unchanged at 2,149 and reaches 0.897 ± 0.020 on the same classifier and held-out boards. Cutting a tight crop around every annotated defect

instead yields 4,274 crops and reaches 0.947, so of the total +0.70 gain, +0.65 is attributable to crop scale—defect-centred cropping using the ground-truth boxes—and +0.05 to the additional crops.

3.2 Synthetic data does not help

Table 1 gives the main result. At 25, 50 and 100%, B falls below A by 0.046, 0.045 and 0.051; a paired test over the three seeds makes the decrease significant at 25% ($p = 0.01$) but not at 50% ($p = 0.06$) or 100% ($p = 0.15$). At 10% the table shows +0.012, which a ten-seed rerun revised to -0.008 macro-F1 (95% CI $[-0.031, +0.014]$; $t(9) = -0.81$, $p = 0.44$), so a benefit larger than +0.014 is unlikely. The harm is consistent with unreliable labels: at best 53% (38/72) of generated crops are assigned by a six-class classifier to the class they were conditioned on. Filtering removes the detectable harm—no $C - A$ difference is significant at any level—but no condition at any level produces a detectable gain.

Table 1 Macro-F1 on held-out boards, mean of three seeds. A: real only; B/C: plus synthetic, unfiltered/filtered. Final column: change from that experiment's own real-only baseline when the generator instead trains on all crops (at 100% this equals B).

Level	A	B	C	$B_{\text{full-pool}} - A$
10%	0.419	0.431	0.433	+0.002
25%	0.641	0.595	0.632	-0.066
50%	0.804	0.759	0.814	-0.017
100%	0.899	0.848	0.884	—

3.3 The generator, not its data, is the bottleneck

Training the generator on all available crops while the classifier stays at 10% buys +0.002 (Table 1, final column), and at 25 and 50% it hurts, so the shortfall is not caused by starving the generator of data. Replacing generated crops with VQ-VAE-2 reconstructions of real ones instead lifts the 10% classifier by +0.51. That comparison adds 2,149 reconstructions against 360 generated crops, so it is an upper bound rather than a like-for-like comparison; the generator comes nowhere near it. Fig. 1 shows why: the samples are plausible board texture with no recognisable defect.

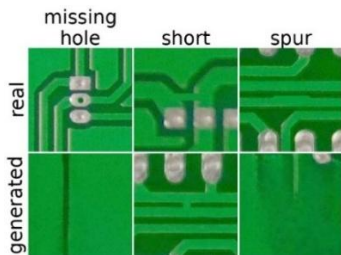


Fig. 1 Real (top) and generated (bottom) tight crops, three of six classes. Samples are a random draw from the full-pool generator's output.

3.4 A defect/no-defect variant agrees

An unconditional generator trained only on defect crops labels every sample a positive by construction, removing class conditioning from the question. On that binary task the gain is +0.018 at the 10% level (95% CI $[-0.058, +0.094]$) and under +0.008 at the other three levels. Its generator also saw crops its classifier did not, which can only have favoured it, yet no benefit appears.

4 Discussion and limitations

The shortfall is therefore not intrinsic to synthetic data but specific to this generator, which is consistent with the field's shift towards external knowledge, such as the pretrained diffusion backbone of [1], over generators fitted to the scarce set alone. Limitations are: one template-synthetic dataset; classification as a proxy for detection; three seeds over classifier initialisation with a single subsample per level; and defect-centred cropping presumes the defect is already localised. These results revise our own earlier qualitative claims [2].

5 Conclusion

VQ-VAE-2 synthetic defect data gives no detectable benefit at any level of data scarcity and lowers mean performance at three of the four tested; training the generator on all available crops does not change this. Reconstructions of real crops are worth +0.51 macro-F1, and defect-centred cropping +0.65, so preprocessing choices warrant examination before generative augmentation. Code and results are at github.com/edwardyapp/pcb-generative-augmentation.

6 References

- [1] Zhang, X., Xu, M., Zhou, X.: 'RealNet: a feature selection network with realistic synthetic anomaly for anomaly detection'. Proc. IEEE/CVF CVPR, Seattle, USA, June 2024, pp. 16699–16708
- [2] Yapp, E.K.Y., Yeh, H.-Y.: 'Generative AI for defect generation in PCBs'. Proc. IEEE ICCE-TW, Taoyuan, Taiwan, July 2026, pp. 923–924
- [3] Razavi, A., van den Oord, A., Vinyals, O.: 'Generating diverse high-fidelity images with VQ-VAE-2', Adv. Neural Inf. Process. Syst., 2019, 32, pp. 14866–14876
- [4] Huang, W., Wei, P.: 'A PCB dataset for defects detection and classification', arXiv:1901.08204, 2019
- [5] Ding, R., Dai, L., Li, G., et al.: 'TDD-net: a tiny defect detection network for printed circuit boards', CAAI Trans. Intell. Technol., 2019, 4, (2), pp. 110–116